BODDU SESHA VENKATA SAI RANGA AKHILESH

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EDUCATION

B. Tech in Electronics and Communication Engineering with Specialisation in Artificial Intelligence & Machine Learning

Deenbandhu Chhotu Ram University of Science and Technology, Murthal (2021 – 2025)

CGPA: 8.32

PROFESSIONAL EXPERIENCE

Project Scientist, Centre for Atmospheric Sciences (CAS) & Transportation Research and Injury Prevention Centre (TRIPC), Indian Institute of Technology Delhi

July 2025 - Present

- Built and deployed an LLM-enabled document ingestion and tagging pipeline (PDF/DOCX/URLs) using FastAPI, Celery and Redis, with robust retries and deduplication via content fingerprinting.
- Implemented semantic search (embeddings) with PostgreSQL and pgvector, along with keyword search using Postgres Full-Text Search, enabling hybrid retrieval for better relevance.
- Built a TypeScript/Node.js backend (using Express & Prisma) to handle document metadata, retrieval APIs, and background workers; integrated secure file upload and parsing workflows.
- Created web-to-text extraction using headless browsing and readability parsing to convert dynamic pages into clean, searchable text reliably.
- Integrated LLM-based Q&A with retrieval-backed context (RAG), including context assembly, chunking, and prompt safety guardrails.
- Built a robust ingestion system using 1400-character chunks with 220-character overlaps and SHA-256 content fingerprinting for deduplication and traceable revisions.

Indian Institute of Technology Delhi | Research Intern in Electronics and Data Analysis using Machine Learning

January 2025 - June 2025

- Engineered and implemented multi-sensor (CO2, Temperature, Humidity, Particulate Matter, Anemometer, Gyroscope, Accelerometer and Geomagnetic) IoT systems by designing circuits, achieving real-time data logging, remote monitoring via web server integration and battery-powered operation.
- Developed and optimised data analysis and visualisation pipelines using Python, employing linear regression, outlier elimination, and smoothing techniques to calibrate sensor measurements.

Centre for Development of Telematics, New Delhi | Electronics Trainee

July 2023 - August 2023

- Grasped the intricacies of programming analog and digital blocks within PSoC microcontrollers, contributing to 2 successful schematic designs that enhanced functionality for LCD applications in integrated environments.
- Prototyped application-specific peripherals utilising hardware description languages.
- Engineered a high-performance LCD controller that cut UI load times by 20% and added metro-style scrolling text + image support, significantly enhancing user engagement by 25%.

SKILLS

SOFTWARE SKILLS:

Python, SQL, HTML, CSS, JavaScript, TypeScript, React, PostgreSQL, Express.js, Prisma, REST API, Celery & Redis Queues, Node.js, pgvector, puppeteer (web extraction), API Security, Docker

AI/ML SKILLS:

Data Analysis/ RAG/ NLP: Machine Learning, Python Libraries (NumPy, Pandas, Scikit-Learn, Keras, Matplotlib, Plotly), Deep Learning, Artificial Intelligence, Statistics and Predictive Analysis, Data Analytics, Natural Language Processing, Large Language Models (OpenAI API), Retrieval Augmented Generation, FastAPI, KeyBERT, YAKE, spaCy, sentence-transformers, embeddings, semantic search, hybrid retrieval (vector and keyword)

Audio ML / DSP: Audio feature engineering, signal processing with librosa/SciPy, windowing/segmentation, Audio transformers(HTS-AT, AST and BERT)

ELECTRONICS SKILLS:

Digital Signal Processing, Internet of Things, Rapid Prototyping, Autodesk Fusion (3D designs), Embedded C, PSoC 5LP (ARM based processors), Raspberry Pi, Arduino (ATMega 328P), Seeed Studio Xiao ESP32S3

PROJECTS

DENSITY-BASED INTELLIGENT TRAFFIC CONTROL SYSTEM

August 2024 - December 2024

- Devised an AI driven system using Raspberry Pi 4, YOLOv10, and OpenCV to adjust traffic signals based on real-time vehicle density dynamically.
- Engineered a real-time object detection pipeline for vehicle counting, integrating computer vision and Pandas for adaptive signal timing and designed a Tkinter-based GUI to visualise traffic density.
- Researched edge-computing techniques for running deep learning models on low-power hardware, proposing future IoT and V2X communication enhancements for smart city integration.

SMART CANE FOR VISUALLY IMPAIRED PEOPLE - AN IIT ROORKEE SPONSORED PROJECT

May 2023 - September 2023

- Redesigned a high-tech mobility aid capable of detecting overhead obstacles, inanimate objects, potholes, and stairs; increased user confidence and mobility by 35% compared to current smart canes through innovative features.
- Launched an innovative feedback system incorporating a buzzer, four vibration motors, and five LED lights, reducing feedback response time by approximately 30%.
- Developed a cost-effective smart cane design using 3D printing, reducing production costs by 10-15%, demonstrating my ability to deliver innovative solutions within budget.

CERTIFICATES

- Introduction of Internet of Things from IIT Kharagpur
- Deep learning from IIT Kharagpur
- Deep learning for Natural Language Processing from IIT Kharagpur
- Computer Vision and Image Processing from IIT Guwahati

ACHIEVEMENTS

- 2ND Position, Tinker and Build in Technova (DCRUST Technical Fest) among 10 teams.
- Won the fellowship of Rs 1 lakh during the State Level Hackathon, organised by DCRUST and IIT Roorkee, recognised as a top solution among 40+ competitors.
- GATE EC 2024 & DA 2025 qualified